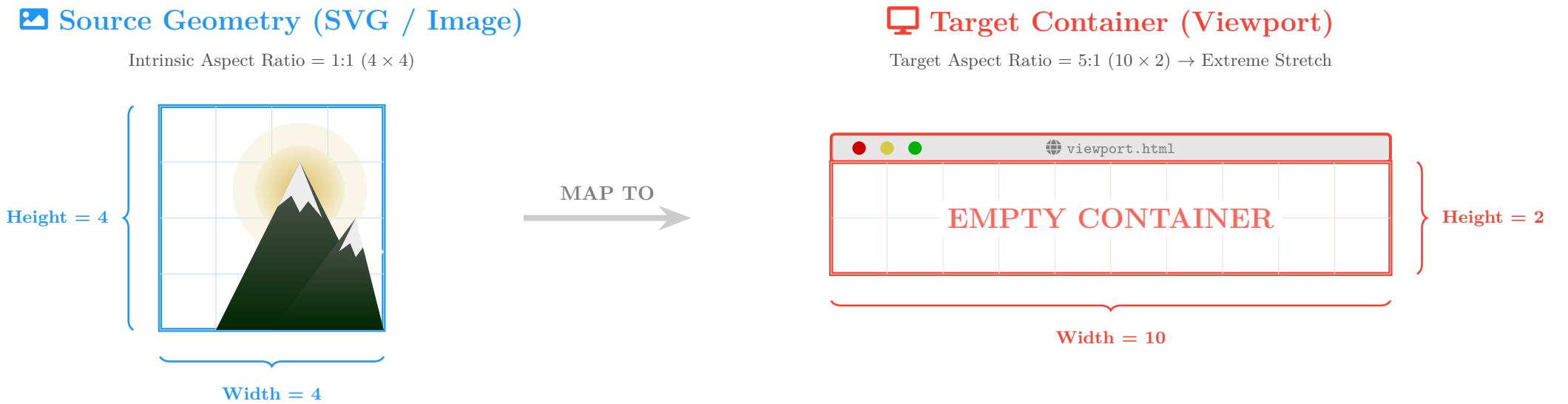


SVG preserveAspectRatio="xMidYMid slice" $\equiv$ CSS object-fit: cover

Structural Equivalence under Extreme Horizontal Stretching



Algorithmic Transformation Pipeline

Both `object-fit: cover` and `preserveAspectRatio="xMidYMid slice"` resolve via identical matrix logic:

- Max Scale Ratio (S):** Ensure the viewport is completely covered without distortion.

$$S_x = \frac{W_{vp}}{W_{src}} = \frac{10}{4} = 2.5, \quad S_y = \frac{H_{vp}}{H_{src}} = \frac{2}{4} = 0.5 \implies S = \max(2.5, 0.5) = 2.5$$

- Dimensional Projection:**

$$W_{scaled} = 2.5 \times 4 = 10, \quad H_{scaled} = 2.5 \times 4 = 10$$

- Centering Translation ($\tilde{T}$):** Align coordinate centers (xMid, yMid / center center)

$$T_x = \frac{W_{vp} - W_{scaled}}{2} = \frac{10 - 10}{2} = 0, \quad T_y = \frac{H_{vp} - H_{scaled}}{2} = \frac{2 - 10}{2} = -4$$

The Geometric Result: Sliced / Cropped

